



higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MECHATRONIC SYSTEMS
NQF LEVEL 2

(6040102)

27 February 2026 (Y-paper)
13:00–16:00

This question paper consists of 6 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA
NATIONAL CERTIFICATE (VOCATIONAL)
MECHATRONIC SYSTEMS
NQF LEVEL 2
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
 2. Read all the questions carefully.
 3. Number the answers according to the numbering system used in this question paper.
 4. Use only a black or blue pen.
 5. Write neatly and legibly.
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2)

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QUESTION 1

- 1.1 Colour plays an important role with regard to input and output information in a technical system.
- 1.1.1 Explain the purpose of colour as an input. (2 × 1) (2)
- 1.1.2 Explain the purpose of colour as an output. (2 × 1) (2)
- 1.2 A bimetal is a device that converts temperature changes into mechanical movement.
- 1.2.1 Briefly explain the construction of a bimetal. (4 × 1) (4)
- 1.2.2 List FOUR applications of a bimetallic strip. (4 × 1) (4)
- 1.3 Explain under the following headings how the bimetal principle works.
- 1.3.1 Heating (2 × 1) (2)
- 1.3.2 Bending (2 × 1) (2)
- 1.3.3 Cooling (2 × 1) (2)
- 1.4 Machines are designed to take one form of energy and convert it by means of a process into another form of energy. These processes can either be mechanical or electrical.
- Name TWO mechanical processes that can take one form of energy and convert it. (2 × 1) (2)
- [20]**

QUESTION 2

- 2.1 List FIVE types of proximity switches. (5 × 1) (5)
- 2.2 2.2.1 Explain by listing FOUR strengths of an optical proximity switch. (4 × 1) (4)
- 2.2.2 Explain by listing THREE weaknesses of the optical proximity switch. (3 × 1) (3)
- 2.3 With reference to mechatronic systems, briefly explain why computers are being used in industry. (3 × 1) (3)
- 2.4 List FIVE disadvantages of using a CAD system. (5 × 1) (5)
- [20]**

2)

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QUESTION 3

3.1 Explain how you would prepare a pneumatic system under the following headings:

- | | | | |
|-------|-------------|---------|-----|
| 3.1.1 | Air supply | (2 × 1) | (2) |
| 3.1.2 | Filtering | (2 × 1) | (2) |
| 3.1.3 | Regulation | (2 × 1) | (2) |
| 3.1.4 | Lubrication | (2 × 1) | (2) |

3.2 3-port valves are used to control the flow of air in the circuit. The ports of the valves are opened and closed by means of a spool.

FIGURE 1 and FIGURE 2 below show the normally closed and the activated 3-port valves.

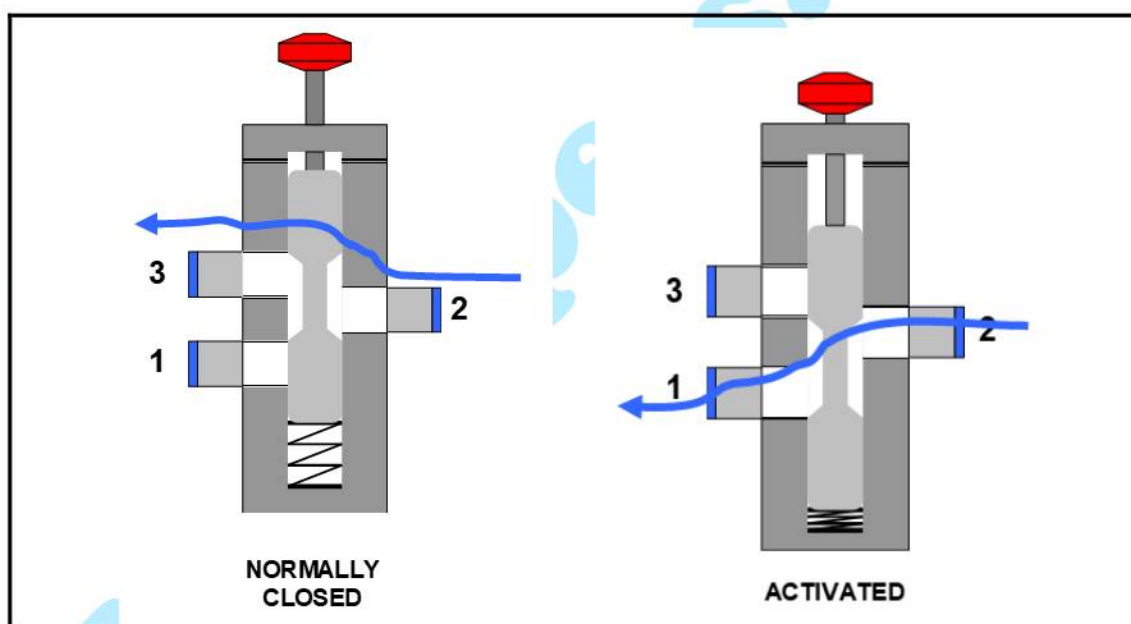


FIGURE 1

FIGURE 2

Explain the difference between a normally closed valve and an activated valve. (3 × 2) (6)

3.3 Draw the symbol for a 5/3-way valve with exhaust ports, pressure port and output ports or working ports. (3 × 1) (3)

3.4 When maintaining or working with pneumatic systems, care must be taken with the removal and connection of the air lines.

Give THREE examples of protection by control and signalling devices. (3 × 1) (3)

[20]



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QUESTION 4

- 4.1 Electricity is commonly used for control signals in remote control applications due to its unique advantages.
- 4.1.1 List FIVE advantages of using electricity for the control signals for remote control applications. (5 × 1) (5)
- 4.1.2 Explain protection by earthing or grounding. (3 × 1) (3)
- 4.2 List FOUR applications where counters are used in electro-pneumatic control. (4 × 1) (4)
- 4.3 Explain with the aid of the given electro-pneumatic control diagram shown below in FIGURE 3, how a cylinder can be retracted after the timer has timed out. (5 × 1) (5)

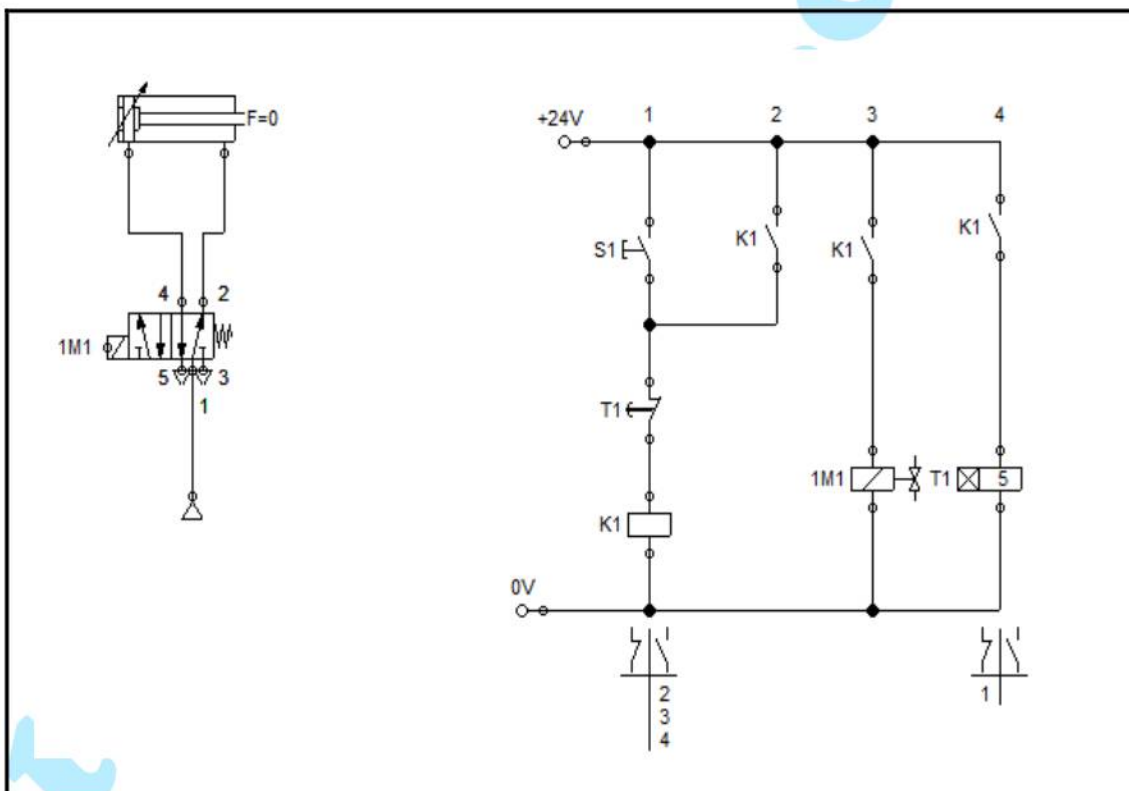


FIGURE 3

- 4.4 Draw the symbol of a normally closed time delay valve. (3 × 1) (3)

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QUESTION 5

- 5.1 Differentiate between fault finding and fault diagnosis. (3 × 2) (6)
- 5.2 The maintenance work can be carried out at short intervals or longer intervals.
List SIX maintenance works that can be carried out at longer time intervals. (6 × 1) (6)
- 5.3 Under the Personal Safety explain precautions to take when performing continuity testing. (3 × 1) (3)
- 5.4 What precautions should be taken when removing tubing from a pneumatic circuit? (2 × 1) (2)
- 5.5 Explain how you would protect a circuit from an over-load current. (3 × 1) (3)

[20]

TOTAL: 100

